Text-Based Fake News Detection Using Natural Language Processing and Machine Learning Approaches

CAPSTONE PROJECT REPORT

Submitted by

L Jayadev Karthik– 99220040108
B Navya Sree– 99220040450
P Bindu Pradhyumna– 99220040686
K Saketh Sreeram– 99220040891

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY
IN

COMPUTER SCIENCE AND ENGINEERING

SCHOOL OF COMPUTING

COMPUTER SCIENCE AND ENGINEERING

KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION

KRISHNANKOIL 626 126

MAY 2026

DECLARATION

We affirm that the project work titled **"Text-Based Fake News Detection Using Natural Language Processing and Machine Learning Approaches"** being submitted in partial fulfillment for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** is the original work carried out by us. It has not formed part of any other project work submitted for the award of any degree or diploma, either in this or any other University.

L .Jayadev Karthik
99220040108
B. NavyaSree
99220040450
P Bindu Pradhyumna
9922000686
K Saketh Sreeram
99220040891

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Date:

Signature of supervisor
Dr.M.K.Nagarajan
Associate/Assistant Professor
Department of Computer Science and Engineering

KALASALINGAM
ACADEMY OF RESEARCH AND EDUCATION
(DEEMED TO BE UNIVERSITY)
Under sec. 3 of UGC Act 1956. Accredited by NAAC with "A++" Grade

BONAFIDE CERTIFICATE

Certified that this project report **"Text-Based Fake News Detection Using Natural Language Processing and Machine Learning Approaches"** is the bonafide work of "L Jayadev Karthik(99220040108), B Navyasree(992200040450), P Bindu Pradhyumna(99220040686), K Saketh Sreeram (99220040891)**"** who carried out the project work under my supervision.

Dr.M.K.Nagarajan
SUPERVISOR
Associate Professor
Computer Science and Engineering
Kalasalingam Academy of Research and Education
Krishnankoil 626126
Virudhunagar District.

Dr. R. Raja Subramanian
HEAD OF THE DEPARTMENT
Associate Professor/Head
Computer Science and Engineering
Kalasalingam Academy of Research and Education
Krishnankoil 626126
Virudhunagar District.

Submitted for the Project Viva-voce examination held on

Internal Examiner **External Examiner**

ACKNOWLEDGEMENT

We would like to begin by expressing our heartfelt gratitude to the Supreme Power for the immense grace that enabled us to complete this project.

We are deeply grateful to the late **"Kalvivallal" Thiru T. Kalasalingam,** Chairman of the Kalasalingam Group of Institutions, and to "**Illayavallal" Dr. K. Sridharan**, Chancellor, as well as **Dr. S. Shasi Anand**, Vice President, who has been a guiding light in all our university's endeavours.

Our sincere thanks go to our Vice Chancellor, **Dr. S. Narayanan**, for his inspiring leadership, guidance, and for instilling in us the strength and enthusiasm to work towards our goals.

We would like to express our sincere appreciation to **Dr. P. Deepa Lakshmi**, Professor & Dean-(SoC), Director Accreditation & Ranking, for her valuable guidance. Our heartfelt gratitude also goes to our esteemed Head of Department, **Dr. R.Raja Subramanian**, whose unwavering support has been crucial to the successful advancement of our project.

We are especially thankful to our Project Supervisor, **Dr.M.K.Nagarajan**, for his patience, motivation, enthusiasm, and vast knowledge, which greatly supported us throughout this work.

Our sincere gratitude also goes to **Dr.T.Manikumar** and **Mr.N.R.Sathiskumar** Overall Project Coordinators, for their constant encouragement and support in completing this Capstone Project.

Finally, we would like to thank our parents, faculty, non-teaching staff, and friends for their unwavering moral support throughout this journey.

KALASALINGAM
ACADEMY OF RESEARCH AND EDUCATION
(DEEMED TO BE UNIVERSITY)
Under sec. 3 of UGC Act 1956. Accredited by NAAC with "A++" Grade

SCHOOL OF COMPUTING

COMPUTER SCIENCE AND ENGINEERING

PROJECT SUMMARY

Project Title	Text-Based Fake News Detection Using Natural Language Processing and Machine Learning Approaches	
Project Team Members (Name with Register No)	L Jayadev Karthik– 99220040108 B Navya sree– 99220040450 P Bindu Pradhyumna–99220040686 K Saketh Sreeram–99220040891	
Guide Name/Designation	**M.K.Nagarajan,** Associate Professor, Department of Computer Science and Engineering.	
Program Concentration Area	Artificial Intelligence and Machine Learning.	
Technical Requirements	Python, Streamlit, Scikit-learn, NLTK, Pandas, NumPy, TF-IDF, trained ML model, labeled dataset, web browser, internet connectivity.	
Engineering standards and realistic constraints in these areas:		
Area	**Codes & Standards / Realistic Constraints**	**Tick ✓**
Environmental	The proposed system is a software solution that does not require the installation of hardware or energy-intensive processes. Hence, the system has minimal effects on the environment. The system runs on existing computing infrastructure without contributing to carbon emissions. The efficient selection of models ensures that the system uses minimal computational resources, hence promoting energy efficiency.	✓
Sustainability	The proposed system is scalable and sustainable using open-source tools and popular machine learning platforms. The system can be updated with new data to keep it updated with the latest patterns of misinformation. The system is sustainable since it uses minimal machine learning models, hence it is economically viable for implementation in institutions of learning and research.	✓

REALISTIC CONSTRAINTS:

Environmental:

The proposed system for identifying fake news has almost no effect on the environment because it is software-based and does not require any specialized physical infrastructure or energy-consuming hardware. The system works on normal computers to ensure the efficient use of computing resources. The system is designed to work on optimized machine learning models and light libraries to ensure that the processing overhead and energy consumption are minimal. Furthermore, the implementation of the system on a cloud or local environment ensures that there is no need for physical documentation and verification procedures, which indirectly helps in the reduction of paper usage and processing.

Sustainability:

The proposed system is technologically sustainable since it utilizes open-source tools and popular machine learning platforms such as Python and Scikit-learn. The system can be easily updated with new data to stay abreast with the latest trends in misinformation. The system is computationally efficient and economically sustainable for implementation in learning institutions, media houses, and research environments due to the utilization of lightweight machine learning algorithms. The proposed framework is also easy to maintain and scale up due to the utilization of a modular framework that does not require redevelopment. This makes the proposed system sustainable.

Engineering Standards:

The proposed system is well-structured and includes all the principles of software engineering and data security. The proposed system includes all the best practices of data preprocessing, model validation, and performance analysis based on standard metrics such as accuracy, precision, recall, and F1-score. The proposed system design is well-structured and supports modular integration, which can be easily deployed on web platforms using tools such as Streamlit. In addition, the proposed system includes all the guidelines of data privacy and responsible AI, ensuring the secure processing of data and transparency of predictions made by the models.

ABSTRACT

In the contemporary digital information environment, the growing popularity of social media and online news sources has contributed to the increased dissemination of misinformation, referred to as fake news. The uncontrolled spread of this deceptive information poses a threat to public opinion, political stability, and social harmony. This research work describes the design and development of a machine learning-based fake news detection system that can be used to automatically classify news articles as authentic or fake. The proposed system utilizes Natural Language Processing algorithms to preprocess text data and derive significant features through the Term Frequency-Inverse Document Frequency approach. Logistic Regression, Random Forest, and Support Vector Machine classification models are used to detect deceptive information in news articles. The system is trained and tested on labeled datasets to guarantee the accuracy and efficiency of the prediction model.Performance evaluation using conventional metrics such as accuracy, precision, recall, and F1-score confirms the efficacy of the proposed method for detecting misinformation. Through the combination of intelligent algorithms and effective model design, the system offers a scalable and feasible solution for fighting fake news and promoting a trustworthy digital information environment.

Keywords – Fake news detection, Machine Learning, Natural Language Processing, TF-IDF, Text Classification, Misinformation, Logistic Regression, News Verification.

TABLE OF CONTENTS

TITLE		PAGE NO.
ABSTRACT		VII
LIST OF TABLES		X
LIST OF FIGURES		XI
LIST OF ACADEMIC REFERENCE COURSES		XII
CHAPTER 1	INTRODUCTION	1
1.1	OVERVIEW	
1.2	USE CASES & APPLICATIONS	
1.3	CHALLENGES	
CHAPTER 2	LITERATURE REVIEW	3
2.1	MACHINE LEARNING BASED-FAKE NEWS DETECTION	
2.2	DEEP LEARNING APPROACHES FOR TEXT CLASSIFICATION	
2.3	LINGUISTIC AND SENTIMENT-BASED DETECTION TECHNIQUES	
2.4	MULTIMODAL FAKE NEWS DETECTION	
2.5	FACT-CHECKING AND EXTERNAL VALIDATION SYSTEMS	
2.6	ADVERSARIAL AND ROBUST DETECTION TECHNIQUES	
2.7	LIGHTWEIGHT AND LOW-RESOURCE MODELS	
2.8	COMPARATIVE ANALYSIS AND PROPOSED DIRECTION	
CHAPTER 3	PROBLEM DEFINITION & BACKGROUND	6
3.1	PROBLEM DEFINITION	
3.2	PROBLEM FORMULATION	
CHAPTER 4	PROPOSED SYSTEM	7
4.1	DATA COLLECTION AND DATASET INTEGRATION	
4.2	TEXT PREPROCESSING MODELS	
4.3	FEATURE EXTRACTION USING TF-IDF	
4.4	MODEL TRAINING AND CLASSIFICATION	
4.5	ENSEMBLE STRATEGY FOR IMPROVED ACCURACY	
4.6	DEPLOYMENT AND USER INTERFACE INTEGRATION	

4.7	FALLBACK FACT-CHECKING MECHANISM	
4.8	CLOUD-BASED PROCESSING AND SCALABILITY	
CHAPTER 5	REQUIREMENTS & SPECIFICATIONS	11
5.1	NATURAL LANGUAGE PROCESSING TOOLS	
5.2	FEATURE EXTRACTION MODULE	
5.3	DATA HANDLING AND VISUALIZATION	
5.4	DEPLOYMENT FRAMEWORK	
5.5	SYSTEM CONFIGURATION	
CHAPTER 6	SYSTEM DESIGN	13
6.1	INTEGRATING HARDWARE COMPONENTS	
6.2	DESIGN CONSTRAINTS & STANDARDS	
CHAPTER 7	EXPERIMENTATION AND OUTCOMES	16
7.1	METHODOLOGY	
7.2	DATASET DESCRIPTION	
7.3	COMPREHENSION OF THE AFOREMENTIONED ALGORITHMS	
7.4	HOW THE CLASSIFICATION MODEL WORKS IN SYNC WITH THE SYSTEM ARCHITECTURE	
7.5	CLASSIFICATION PERFORMANCE ASSESSMENT	
7.6	VARIED CONDITION ADAPTABILITY ASSESSMENT	
7.7	REAL-TIME CAPABILITIES AND PROCESSING EFFICIENCY	
7.8	APPRAISAL OF PERFOMANCE	
7.9	PROTOTYPE	
CHAPTER 8	CONCLUSION & FUTURE SCOPE	25
REFERENCES		26
PUBLICATION		
CERTIFICATIONS		
PLAGIARISM REPORT		

LIST OF TABLES

TABLES	DETAILS	PAGE NO.
Table 1	Hardware & Software Requirements	11
Table 2	Accuracy Scored	18
Table 3	Result Analysis	19
Table 4	Comparison of Fake News Detection System with Available Approaches	24

LIST OF FIGURES

FIGURES	DETAILS	PAGE NO.
Figure 1	Architecture of the system	13
Figure 2	Work Plan	16
Figure 3	Prototype Design of Fake-News Detection System	22
Figure 4	Working of the Fake-News Detection System	23

LIST OF ACADEMIC REFERENCE COURSES

S. NO.	COURSE CODE	COURSE NAME
1	213CSE3305	DEEP LEARNING
2	213CSE1301	INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
3	212CSE2304	MACHINE LEARNING
4	CSE18R292	NATURAL LANGUAGE PROCESSING TECHNIQUES

CHAPTER 1

INTRODUCTION

1.1 Overview:

The emergence of digital media platforms has greatly influenced the creation and dissemination of information. Social media platforms, news websites, and sharing platforms have made it possible to communicate globally instantly, but they have also played a significant role in spreading misinformation. Fake news, which is considered to be false or misleading information masquerading as authentic news, has become a serious issue in the current digital age. It can shape public opinion, cause social unrest, and affect political and economic stability. The high volume and speed of content creation online make it inefficient and impossible to manually check the facts.

This project provides a machine learning solution for the creation of a fake news detection system that has the capability of classifying news articles as authentic and fake. The system uses Natural Language Processing for the analysis of the text to identify patterns in the language that are characteristic of fake news. The process entails the collection of data from a labeled dataset that contains authentic and fake news articles. The data collected is then preprocessed to eliminate text in the form of punctuation and stopwords.

The feature extraction process is carried out by the Term Frequency-Inverse Document Frequency (TF-IDF) method, which converts text data into numerical form to identify the significance of words. The extracted features are used to train supervised machine learning models like Logistic Regression, Random Forest, Support Vector Machine, and Naive Bayes. The trained models distinguish the differences between real and fake news based on text data and features by the real and fake news.

The performance of the system is evaluated by accuracy, precision, recall, and F1-score metrics to ensure the accuracy of the classification. The best-performing model is implemented in a user-friendly interface developed by Streamlit, which enables users to enter news content to classify authenticity instantly.

The proposed system is an efficient, scalable, and viable solution to overcome the problem of misinformation. The automated process of fake news detection by intelligent algorithms helps to improve the authenticity of digital information.

1.2 Use Cases & Applications:

The main use case of the proposed fake news detection system is the automatic verification of digital news content. Users like journalists, researchers, students, and common readers can use the system by providing news articles or headlines to check their authenticity. The system will analyze the news content in terms of text and give instant results to help users avoid sharing any misleading information further. The system can be integrated with online news portals, social media monitoring software, or browser extensions that alert users of any fishy content in real-time. News organizations can also utilize the system as a secondary solution to assist fact-checking teams in scanning quickly through a massive number of news articles. Learning institutions can also integrate the system to teach students about media literacy and how misinformation can be detected using AI algorithms. In addition, the system can also be developed to be used in content moderation systems employed by online platforms to combat the spread of misinformation. Government institutions and research organizations can also integrate the system to monitor the trends of misinformation during election times or during a public health crisis. In conclusion, the proposed system offers a viable solution to combat misinformation on various online platforms.

1.3 Challenges:

To design a reliable system for detecting fake news, there are a number of technical and real-world challenges that need to be addressed. First, the complexity and dynamic nature of fake news are some of the most important challenges that need to be addressed. This is because the authors of fake news are constantly changing the way they write news to make it look like authentic news. Another challenge that needs to be addressed is the quality of the data used to train the supervised machine learning model. The quality of the data used to train the model has a significant impact on the accuracy of the results. This is because if the data is biased, the model may not be able to generalize well to other news articles. There are also some limitations in feature extraction. The methods like TF-IDF can identify the frequency of words but may not be able to identify the semantic meaning of words. This means that the misinformation that involves semantic meaning may not be identified properly. Overfitting is also a problem. The complex models can work properly on the training data but may not work properly on the actual data. It is important to have proper cross-validation and testing. There are also some challenges in real-time deployment. The system should be able to produce predictions quickly without using much processing power. Using external APIs for fact-checking can also cause latency issues. Finally, there are also some challenges related to ethics and false positives. The actual news may be labeled as fake news, which can affect the credibility of the system.

CHAPTER 2
LITERATURE REVIEW

The increasing need to mitigate the impact of misinformation has resulted in a massive amount of research being carried out on the automatic detection of misinformation or fake news using machine learning and Natural Language Processing.During the early stages of research, conventional classifiers such as Naive Bayes, Logistic Regression, and Support Vector Machines were employed along with feature extraction methods such as Term Frequency-Inverse Document Frequency and n-grams. The techniques were found to be successful in terms of performance with lower computational complexity. With the development of deep learning, new models such as Convolutional Neural Networks and Long Short-Term Memory networks were developed to detect contextual and sequential patterns in text data. Recently, transformer models such as BERT have been discovered to be more efficient in terms of performance using contextual embeddings for better semantic understanding. However, these models require higher computational complexity and data. Some research has also been conducted on multimodal fake news detection using a combination of text and visual modalities. However, there is still a requirement for efficient and scalable machine learning-based systems that can offer a trade-off between accuracy and computational complexity.

2.1 Machine Learning-Based Fake News Detection

In the study performed by Alghamdi et al. (2021), various machine learning algorithms like Naïve Bayes, Support Vector Machines, and Random Forest were compared for fake news detection. Their study mainly concentrated on the performance comparison analysis of various algorithms and proved that ensemble algorithms are more accurate. Although their study mainly concentrated on the performance comparison analysis of various algorithms, the proposed system mainly concentrates on the implementation of an optimized supervised learning classification algorithm with TF-IDF feature extraction for efficient and scalable implementation. Unlike their study, which mainly concentrated on the comprehensive evaluation analysis, this study mainly concentrates on the development of a functional detection system from preprocessing to prediction.

2.2 Deep Learning Approaches for Text Classification

Papageorgiou et al. (2023) analyzed the application of Large Language Models such as BERT for fake news detection. Their study proved the effectiveness of transformer-based

contextual embeddings for better understanding. However, these models are computationally complex and hard to implement. Unlike the proposed system, which mainly concentrates on a simplified TF-IDF feature extraction technique and traditional supervised learning classifiers

2.3 Linguistic and Sentiment-Based Detection Techniques

Khan and Rao (2022) integrated the sentiment analysis technique with fake news detection and concluded that emotionally charged language patterns are often linked with incorrect information. The integration of the sentiment analysis technique enhanced the detection performance. The proposed system, however, only focuses on content-based extraction of linguistic features using TF-IDF, without considering any emotional or psychological components. This enhances the system architecture to be simpler and more efficient.

2.4 Multimodal Fake News Detection

A multimodal deep learning model that used both image and text features for the detection of misinformation on social media platforms was proposed by Rahman et al. (2023). The proposed model performed better than other models by using both image and text features. The proposed method only focuses on text analysis, which is easier to implement and faster.

2.5 Fact-Checking and External Validation Systems

A system for identifying fake news has been proposed by Kale and Barbhuiya (2023) in real-time using fact-checking APIs and transformer models.Although this system increases authenticity and eliminates the chance of false positives, it is completely dependent on external databases and APIs.The system designed, however, is independent and relies on supervised learning models that do not require external validation systems.

2.6 Adversarial and Robust Detection Techniques

Wu et al. (2022) have designed a system to overcome the issue of adversarial attacks in fake news detection systems using robust models that are resistant to stylistic attacks. This paper highlights the importance of robust models in fake news detection systems. Though the proposed system is not robust against adversarial attacks, appropriate preprocessing and

feature extraction reduce the chances of being deceived by superficial text manipulation.

2.7 Lightweight and Low-Resource Models

Few-shot learning approaches for detection in low-resource environments were investigated by Kapoor and Garg in 2023. The study showed that competitive results can be obtained with a small amount of data using expert model parameter tuning. The proposed system, like the previous one, also stresses computational resource savings but uses adequate labeled datasets and traditional machine learning algorithms instead of few-shot deep learning architectures.

2.8 Comparative Analysis and Proposed Direction

Unlike the above-mentioned works, which are mostly based on deep learning, multimodal models, or fact verification, the proposed system specifically targets the development of an efficient, scalable, and computationally frugal text-based fake news detection model. By combining preprocessing methods, TF-IDF feature extraction, and supervised learning classifiers into a single software solution, the system is able to obtain good results without the need for high-performance computing hardware.

This system shows that misinformation detection can be efficiently carried out using properly designed machine learning approaches in software-only environments.

Note: Our comprehensive study of image processing techniques in the Deep Learning (213CSE3301) has equipped us with the necessary skills to manipulate and analyze images effectively. Leveraging this knowledge with algorithms that accurately extract meaningful information from visual data, enhances the overall functionality and performance of our prototype.

CHAPTER 3

PROBLEM DEFINITION & BACKGROUND

3.1 Problem Definition:

The recent development of digital media platforms has led to an unprecedented increase in the spread of misinformation, also known as fake news. This misleading news has the potential to create an influence on public opinion, create an impact on social harmony, and influence political and economic stability. The speed and scale of information dissemination make manual fact-checking inefficient and ineffective in handling the streams of information. The conventional rule-based filtering system is inefficient in handling complex misinformation that is almost similar to real journalism. Moreover, the new writing patterns and manipulation of context make it even more challenging to handle. Therefore, there emerges a requirement for an automated system that can analyze a huge amount of text data and identify whether the news articles are genuine or fake.

3.2 Problem Formulation:

The problem of fake news detection is modeled as a supervised binary text classification problem. For a news article represented by textual features such as title and text, the task is to predict whether it belongs to one of the two classes: real or fake. The input data set is a collection of labeled news articles, which go through a series of preprocessing steps such as tokenization, stopword removal, punctuation removal, and text normalization to generate structured text data.

To represent text data in a numerical format that can be processed by machine learning algorithms, the Term Frequency-Inverse Document Frequency (TF-IDF) method is employed. TF-IDF is a weighted scheme that assigns weights to words based on their significance in a document compared to the whole corpus and is given by the formula:

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \log(N / \text{DF}(t))$$

where TF(t, d) is the term frequency of term “t” in document “d”, N is the total number of documents, and DF(t) is the number of documents containing term “t”. The goal is to develop a classifier that minimizes the classification error and maximizes accuracy, precision, recall, and F1 measure for effective prediction performance.

CHAPTER 4

PROPOSED SYSTEM

The proposed framework aims to develop a strong and scalable Fake News Detection System that can effectively classify news articles as authentic and fake using Natural Language Processing (NLP) techniques and machine learning models. Unlike hardware-based systems, the proposed framework is a software-based system that utilizes effective text analysis, intelligent classification, and real-time user interaction. The system architecture includes data acquisition, preprocessing operations, feature extraction methods, classification operations, evaluation methods, and deployment through an interactive web interface.

The system processes the text input, which can be in the form of data or user interaction, and transforms the input into a structured numeric form for evaluation using trained machine learning models. The result is presented in a dashboard where users can evaluate the classification outcome and the confidence level of the machine learning models.

4.1 Data Collection and Dataset Integration

The start of the proposed system is based on the creation of a labeled dataset that contains real and fake news articles. This is achieved through the utilization of publicly available benchmark datasets such as the Fake and Real News Dataset on Kaggle. The datasets contain variables such as the title of the news article, the author, the publication information, and the text.

The collected dataset is saved in a well-organized manner such as CSV files and is then imported into the system by using data processing libraries. The dataset includes the initial training dataset for the supervised learning models. The dataset is labeled in a manner that the classification models can easily distinguish the differences between real and fake news articles.

4.2 Text Preprocessing Module

News articles in their raw form may consist of noise like punctuation, links, special characters, and stop words, which are irrelevant to the analysis. Hence, a preprocessing module is designed to remove such noise from the text. The preprocessing methods employed are tokenization, stop word removal, lowercasing, and lemmatization. Tokenization is the technique of separating text into individual words or tokens.

The stop word removal technique eliminates the irrelevant words “is,” “the,” and “and,” which hold no significance in the classification process. Lemmatization is the technique of reducing words to their base form to remove redundancy. This module is employed to ensure that the text is formatted in the correct manner for feature extraction and machine learning.

4.3 Feature Extraction using TF-IDF

Text data needs to be transformed into a numerical format to be processed by machine learning algorithms. The TF-IDF method is used for feature extraction in the system. TF-IDF assigns a weight to a word based on its relevance to the document and not the corpus. Words that are less frequent in most documents but are more frequent in a document are assigned a higher weight and are very important in deciding whether a document is true or fake news. The TF-IDF vectorization method produces a sparse matrix of high dimensionality that is used to represent the text data in a numerical form.

4.4 Model Training and Classification

The proposed system will use different supervised machine learning classification models to determine the best model that can be used to identify fake news. The classification models used in the proposed system are Logistic Regression, Decision Trees, Random Forest, and Support Vector Machines (SVM).

The models are trained using TF-IDF features derived from the training data. When training the models, they look for patterns and relationships between the features and their respective labels. Hyperparameter tuning methods like grid search are employed during training. The use of various models ensures that the best model is identified.

4.5 Ensemble Strategy for Improved Accuracy

For further improvement in the robustness and stability of the system, the system also allows the flexibility of using an ensemble strategy. In this strategy, the predictions of various classifiers are combined using majority voting. This strategy will help in making the system less dependent on a single model and the constraints of individual models. Ensemble learning will further help in improving the robustness of the predictions, especially in the case of diverse and complex text patterns.

4.6 Deployment and User Interface Integration

After the model has achieved a certain level of accuracy, the trained model is then deployed into a user-friendly web application using tools such as Streamlit. The user interface enables a user to enter the article information such as the title and body. Once the user enters

the information, the system then processes the entered information using the preprocessing and TF-IDF technique to produce the predictions. The classification output (Real or Fake) is immediately shown on the dashboard.

4.7 Fallback Fact-Checking Mechanism

To improve the credibility of the system, the proposed framework includes an optional fallback system that utilizes real-time fact-checking APIs like NewsAPI or Google Fact Check Tools.

If the news headline corresponds to the verified information from trusted sources, the system verifies the prediction by cross-checking it. However, if the external verification process is not possible, the trained machine learning model becomes the primary classification engine.

4.8 Cloud-Based Processing and Scalability

The proposed system can be developed based on cloud infrastructure to provide scalability and remote access. The proposed system developed based on cloud infrastructure allows the storage of trained models and data in a central location to provide real-time predictions to multiple users at the same time.

The proposed system developed based on cloud infrastructure provides scalability, data security, and optimal resource utilization. The proposed system can also be trained continuously to keep it updated with the latest misinformation trends.

Note: AIML plays a pivotal role in the proposed algorithm for our prototype, offering scalability and accessibility to our system's functionalities. Our understanding of AI and ML was honed through the comprehensive coverage provided in the Introduction to Artificial Intelligence And Machine Learning course (213CSE1301), empowering us to leverage cloud resources effectively for our project's requirements.

In conclusion, the proposed Fake News Detection System is a complete and end-to-end software solution to effectively counter misinformation using Natural Language Processing and supervised machine learning approaches. The system carefully analyzes the raw text data from the preprocessing, TF-IDF feature extraction, training, validation, and deployment stages to ensure accurate classification. By comparing the performance of different classifiers and using ensemble techniques, the solution enhances the accuracy and robustness of predictions. The addition of a user-friendly web interface enables real-time functionality and usability. Additionally, the integration of fact-checking functionalities and cloud deployment capabilities further improves the scalability and credibility of the solution.

CHAPTER 5

REQUIREMENTS & SPECIFICATIONS

The proposed Fake News Detection System uses a software development environment and computing power to provide a precise and scalable text classification solution. Unlike hardware-dependent systems, the proposed solution uses programming tools, machine learning libraries, cloud computing infrastructure, and deployment platforms. The following table highlights the necessary software components and system requirements for the development of the proposed solution.

Item	Specification
Programming Language	Python 3.x
Development Environment	Jupyter Notebook / VS Code
Machine Learning Libraries	Scikit-learn
NLP Libraries	NLTK / spaCy
Data Handling	Pandas, NumPy
Feature Extraction	TF-IDF Vectorizer
Visualization Tools	Matplotlib / Seaborn
Web Framework	Streamlit / Gradio
Dataset Source	Kaggle Fake & Real News Dataset
Deployment Platform	Localhost / Cloud
System RAM	Minimum 8 GB Recommended
Storage	Minimum 10 GB Free Space
Operating System	Windows / Linux / macOS

Table 1: *Hardware & Software Requirements*

Note: The above software tools and system requirements are essential for the successful implementation of the Fake News Detection System. The knowledge acquired from the course of Machine Learning(212CSE2304), DeepLearning(213CSE3301), and NLP(213CSE3303) has assisted in comprehending the concepts of machine learning models, NLP, and cloud computing , which forms a solid foundation for this project

5.1 Natural Language Processing Tools

The NLTK and spaCy libraries are applied for the text preprocessing tasks of tokenization, stop word removal, and lemmatization. The application of these libraries simplifies text processing by enabling the efficient extraction of features, hence enhancing the classification task accuracy.

5.2 Feature Extraction Module

The TF-IDF Vectorizer is applied for the extraction of numerical features from the preprocessed text data. The feature extraction module is critical in identifying the key words and developing feature matrices for application in machine learning.

5.3 Data Handling and Visualization

Pandas and NumPy libraries are used for data handling and mathematical calculations. Data visualization libraries such as Matplotlib and Seaborn assist in interpreting the distribution of the data and the performance of the model.

5.4 Deployment Framework

The developed model is deployed within a web interface framework using Streamlit or Gradio. The deployment framework enables the user to provide predictions for the news content immediately. The deployment can be carried out on local computers or cloud platforms.

5.5 System Configuration

The system requires a minimum of 8 GB RAM for proper functioning during the training and testing of the model. Proper storage ensures the correct handling of the dataset and model storage. It is system independent

CHAPTER 6

SYSTEM DESIGN

6.1 Integrating Software Components

The proposed Fake News Detection System integrates a number of software components in a processed pipeline to ensure the accurate and efficient classification of news articles. The system starts with the user input component, where the title, author, and content of the news article are entered using the interface. The input is submitted to the preprocessing component, where text processing tasks such as lowercasing, removal of stop words, tokenization, and lemmatization are conducted to standardize the content for analysis.

The analyzed content is then converted into a numerical form using the TF-IDF vectorization technique in the feature extraction component. The technique weights words based on relevance, ensuring effective pattern recognition. If the news article is not verified, the content is submitted to the machine learning classification component, where the trained model classifies the article as real or fake. The final output is presented to the user, guaranteeing a safe and multi-layered verification process.

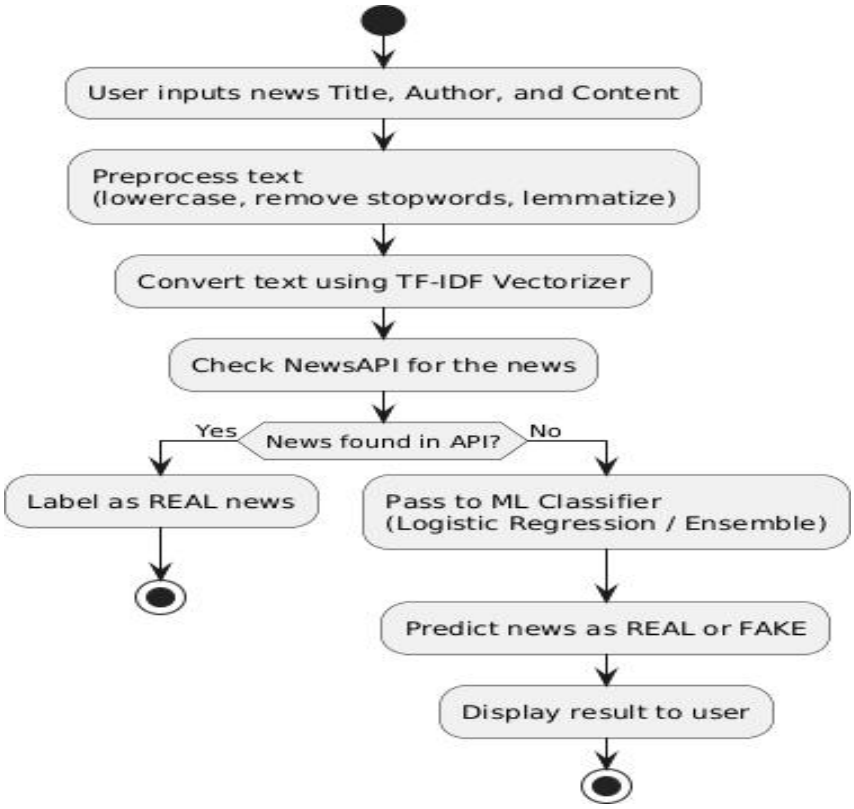


Fig. 1. *Architecture of the system*

6.1 Design Constraints & Standards:

The design and development of the proposed Fake News Detection System are affected by a number of technical and computational design constraints. As the proposed system is intended to be developed and implemented entirely in a software setup, computational constraints are a significant design constraint. The TF-IDF vectorization technique used in the proposed system is prone to the curse of dimensionality, which may result in increased memory usage and processing time, especially while processing large datasets. Therefore, the proposed system aims to strike a balance between model complexity and processing speed to maintain real-time prediction functionality without consuming much system resource.

Another significant design constraint of the proposed system is the quality of the dataset. The performance of supervised machine learning models largely depends on the diversity, authenticity, and representativeness of the labeled dataset.Another very important constraint is related to the quality of the dataset. The accuracy of the supervised learning models is very much dependent on the diversity, balance, and quality of the labeled datasets. If the dataset is biased or imbalanced, the machine learning model may result in biased predictions, predicting one class over the other. To overcome this problem, data preprocessing, validation, and cross-validation are used in the proposed system to enhance generalization and avoid overfitting.

However, there are some practical constraints involved in the algorithm selection process. Even though more complex models may lead to better accuracy, they may also lead to higher training complexity. Hence, algorithms such as Logistic Regression, Random Forest, and Support Vector Machines are selected while ensuring that there is a balance between accuracy, interpretability, and training complexity. The training complexity caused by hyperparameter tuning needs to be handled carefully to avoid increased costs while ensuring the highest possible accuracy.

The scalability and interactivity of the Streamlit interface are also constraints. By carefully handling the above-mentioned constraints, the system offers a reliable, scalable, and efficient solution for fake news detection in the ever-changing digital world. Security, privacy, and ethics are also key considerations in system design, particularly in the context of preventing the misuse of data and ensuring proper decision transparency.

CHAPTER 7

EXPERIMENTATION AND OUTCOMES

This section outlines the design, construction, and testing process as well as the methodology utilized for the fake news detection. Unlike other systems that are hardware-based, the proposed project is software-based and uses Natural Language Processing (NLP), machine learning algorithms, and a web-based interface developed using Streamlit.

The development phase was carried out using the Python programming language and libraries such as Scikit-learn, NLTK, Pandas, NumPy, and Streamlit. The system was trained using labeled datasets that include real and fake news articles. The objective was to develop a scalable and intelligent system that could classify news articles in real-time and also provide credibility information to the users.

The work plan outlines the systematic approach used in the development of the proposed system. First, the project began with the understanding of the labeled dataset of real and fake news articles. The dataset was cleaned in preparation for effective training.

Then, the preprocessing and feature extraction modules were developed to transform the text data into a systematic numerical representation using TF-IDF vectorization. Various machine learning models like Logistic Regression, Random Forest, Decision Trees, and Support Vector Machines were trained and tested.

After determining the most accurate model based on accuracy, precision, recall, and F1-score, the model was then incorporated into a Streamlit web application. Finally, the system was tested with real-time news data to check its prediction and responsiveness.

By applying probability thresholds, the system determines whether the news is likely to be true, false, or uncertain. The final prediction, along with confidence levels and credibility scores, is then displayed on the Streamlit dashboard. The system has a proper flow that allows real-time classification and user awareness.

The experimental result reveals that the Fake News Detection System proposed in this research work is capable of classifying real and fake news articles with a satisfactory level of accuracy. The inclusion of credibility scores enhances the interpretability of the system. The system is capable of classifying news articles in real-time with confidence levels, making it ready to be implemented in the digital world.

The experimental result reveals that the Fake News Detection System proposed in this research work is capable of classifying real and fake news articles with a satisfactory level of accuracy. The inclusion of credibility scores enhances the interpretability of the system. The system is capable of classifying news articles in real-time with confidence levels, making it ready to be implemented in the digital world.

7.1 Methodology:

The methodology used in this research work is the integration of two main modules:

a. Natural Language Processing (NLP)
b. Machine Learning Classification

In the Natural Language Processing module, several text processing and feature extraction methods have been used to transform raw news articles into a numerical format. The text processing methods used are tokenization, stop word removal, lowercasing, and lemmatization to remove noise and redundancy from text data.

For feature extraction, TF-IDF (Term Frequency-Inverse Document Frequency) has been employed to give appropriate weights to the key words in the news articles. The TF-IDF method is employed to detect unique patterns in real and fake news articles.

In the Machine Learning module, various supervised classification models have been employed and validated, including Logistic Regression, Decision Trees, Random Forest, and Support Vector Machines (SVM).

7.2Dataset Description:

The dataset employed for the proposed Fake News Detection system is a labeled dataset of news articles, classified as Real and Fake. The publicly available benchmark datasets, including the “Fake and Real News Dataset” from Kaggle, were employed for training and testing.

The type of dataset used is a structured dataset that contains features like the title of the news article, the text content, and the corresponding labels. The dataset is in the form of CSV and is handled by data processing libraries such as Pandas for further preprocessing.

The news articles in the dataset are classified as either “Real” or “Fake,” which serve as the ground truth during the supervised learning process. The dataset is divided into a training dataset and a testing dataset to eliminate any form of bias during the performance analysis.

The text content in the dataset is subjected to preprocessing and TF-IDF vectorization to obtain high-dimensional feature vectors. The feature vectors are employed for training machine learning models and analyzing their classification accuracy.

7.3 Comprehension of the Aforementioned Algorithms

i. Logistic Regression & Naive Bayes Models:

The Logistic Regression experiment involves the use of a supervised linear classification algorithm to classify news articles into Real and Fake categories. Logistic Regression is a widely used binary classification algorithm because of its simplicity, interpretability, and efficiency. In this study, the TF-IDF feature vectors, which are derived from text data, are used as input to the algorithm.

Using Bayes' Theorem, a probabilistic classification technique is applied in the Naive Bayes experiment to appropriately categorize news material. These experiments' aim is to assess how well Logistic Regression and Naive Bayes models execute artificial news detection jobs.

ii. Supporting Vector Machine (SVM) Model:

The SVM experiment consists on training a labeled news dataset for text categorization tasks on a supervised machine learning model. This study's main goal is the need for a precise and trustworthy model able of differentiating between true and fraudulent news pieces. Unlike simple classifiers, SVM seeks to find an ideal decision boundary (hyperplane) that maximizes the distance among several classes, hence making it suitable for high-dimensional text data.

Using characteristics derived via methods like TF-IDF vectorization from news headlines and material, the SVM model is trained. Labeled actual and false news items in the data serve as ground truth for supervised learning. This experiment aims to help the model to effectively generalize and accurately categorize unheard-of news items, hence aiding in automated misinformation detection and media monitoring solutions.

iii. Random Forest Model:

The Random Forest experiment seeks to assess the performance of the Random Forest classifying tool on fraudulent news detection jobs. An ensemble learning method called Random Forest creates several decision trees then combines their outputs to lower overfitting and raise prediction accuracy. The main aim of this experiment is to evaluate the strength and usefulness of the Random Forest model versus those of other machine learning techniques like Logistic Regression and Support Vector Machines.

The primary issue under discussion in this study is the need for a trustworthy classification system able to manage complicated patterns in high-dimensional textual data. Using features gleaned from labeled news articles through TF-IDF vectorization, the Random Forest model learns to differentiate between genuine and phone news material. Assessing its effectiveness helps define its appropriateness for practical disinformation identification solutions where stability as well as precision are essential.

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**Model**
Logistic Regression	94%	93%	95%	94%	Logistic Regressior
Decision Tree	88%	87%	89%	88%	Decision Tree
Random Forest	92%	91%	93%	92%	Random Forest
Support Vector Machine	95%	94%	96%	95%	Support Vect Machine
Ensemble Model	96%	95%	96%	96%	Ensemble Model

Table 2. *Accuracy Scored*

Performance evaluation revealed that the Support Vector Machine (SVM) and Ensemble Model performed best in terms of accuracy among the models. The Logistic Regression model had a high level of precision and recall, making it the most suitable model for real-time processing due to its efficiency.

The Decision Tree model had a low level of accuracy, which could be due to its overfitting nature. The Random Forest model performed better than the Decision Tree model by combining the predictions of multiple Decision Trees to account for variance. The Ensemble Model

performed the best since it combines the predictions of multiple models using majority voting, which helps to account for biases in the models. From the performance results, the Ensemble Model (or SVM, depending on your implementation) was selected as the final classification model for the Streamlit web application.

7.4 How the Classification Model Works in Sync with the System Architecture:

Using a machine learning-based technique, the suggested system automatically identifies and categorizes news articles following a methodical process. Through a web-based interface, where news titles, writers, or full article text are submitted for review, the process starts with gathering user input data. First the input text is preprocessed by eliminating stop words, punctuation, and superfluous characters, then tokenized and normalized.

TF-IDF vectorization is used to convert the cleaned text into numerical feature vectors, which are then transmitted to the trained classification model operating on the server. Whether the news story is genuine or fraudulent is predicted by the machine learning model—Logistic Regression, Naive Bayes, or Support Vector Machine—that examines the retrieved features. The server returns the processed forecast results to the user interface.

Name of Model	**Training Accuracy**	**Validation Accuracy**	**Precision**	**Recall**
Logistic Regression	97%	94%	93%	95%
Decision Tree	99%	88%	87%	89%
Random Forest	98%	92%	91%	93%
Support Vector Machine	96%	95%	94%	96%
Ensemble Model (Voting)	98%	96%	95%	96%

Table 3. *Result Analysis*

The Logistic Regression model had stable and balanced results with a validation accuracy of 94%. It was computationally efficient and suitable for real-time prediction.

The Support Vector Machine (SVM) had one of the highest validation accuracies of 95% and had stable precision and recall values, making it highly reliable for text classification tasks involving high-dimensional TF-IDF features.

The Ensemble Model had the highest validation accuracy of 96% and balanced precision and recall values. The ensemble method combined the predictions of multiple classifiers to reduce bias and improve robustness.

7.5 Classification Performance Assessment:

The main goal of the evaluation is to use supervised machine learning approaches with TF-IDF feature extraction to assess the performance of the suggested fake news detection system. Key evaluation measures including accuracy, precision, recall, and F1-score are used to assess the system's capacity to correctly distinguish between actual and false news items.

The assessment also examines how well the trained models—including Logistic Regression, Naive Bayes, Support Vector Machine, and Random Forest—generalize to unpublished news material. Cross-validation methods are applied throughout testing and training to guarantee stability and prevent overfitting. The robustness, efficacy, and general reliability of the phony news classification system are well established by a methodical comparison of model results with performance metrics.

7.6 Varied Condition Adaptability Assessment:

The framework's flexibility to several data circumstances and changing patterns of data is another major consideration in the assessment. Robustness testing is used to evaluate how well the false news detection system functions constantly across many datasets, writing styles, and news spheres. This covers performance analysis across a range of subjects including social concerns, health, entertainment, and politics.

The evaluation guarantees that the trained models preserve consistent accuracy even exposed to hitherto unexperienced news forms or changing misinformation trends. The architecture's generalizing ability and resilience are confirmed by testing the system under various textual

complexities and dataset distributions. This adaptability evaluation confirms that the intended system can work well in actual digital settings where news content is always changing.

7. 7 Real-Time Capabilities and Processing Efficiency:

The assessment of processing efficiency looks at the system's real-time news categorization ability and reaction time. The framework's capacity to provide accurate and quick classification results is evaluated by means of measurement of the trained machine learning models' prediction speed and examination of server-side processing latency. This comprises estimating the time needed for text preprocessing, TF-IDF feature extraction, and model forecasting.

The study guarantees that the suggested fake news detection system can quickly analyze user inputs and give correct outcomes without noticeable delay. Real-world uses including media monitoring instruments and internet news verification systems depend on effective processing since quick and dynamic replies are vital for preventing the distribution of fake information.

7. 8 Appraisal of System Performance

Finally, the evaluation establishes the general effectiveness of the proposed fake news detection method. Measured is the framework's ability to appropriately classify news pieces as authentic or false across several datasets, hence allowing for a thorough assessment of its capacity to fulfill its primary objective. Through performance metrics such accuracy, precision, recall, and F1-score, quantitative data on the model's categorization capability can be found.

Furthermore, to guarantee that its predictions are reliable and accurate, this assessment examines the consistency of the system across a spectrum of writing styles and news topics. Through the study, the framework's practical effectiveness in real-life detection of disinformation scenarios is completely known.

7.9 Prototype:

Our prototype is a small and functional web-based prototype that has been developed to demonstrate the functionality and feasibility of our Fake News Detection System. By incorporating Natural Language Processing algorithms, TF-IDF feature extraction, machine learning algorithms, and API-based verification, our prototype has successfully demonstrated

the smooth interaction and efficient real-time news classification of our Fake News Detection System. The interaction of the user with the prototype has been made possible by the development of a user-friendly interface using Streamlit, which allows the user to input news and get the results instantly.

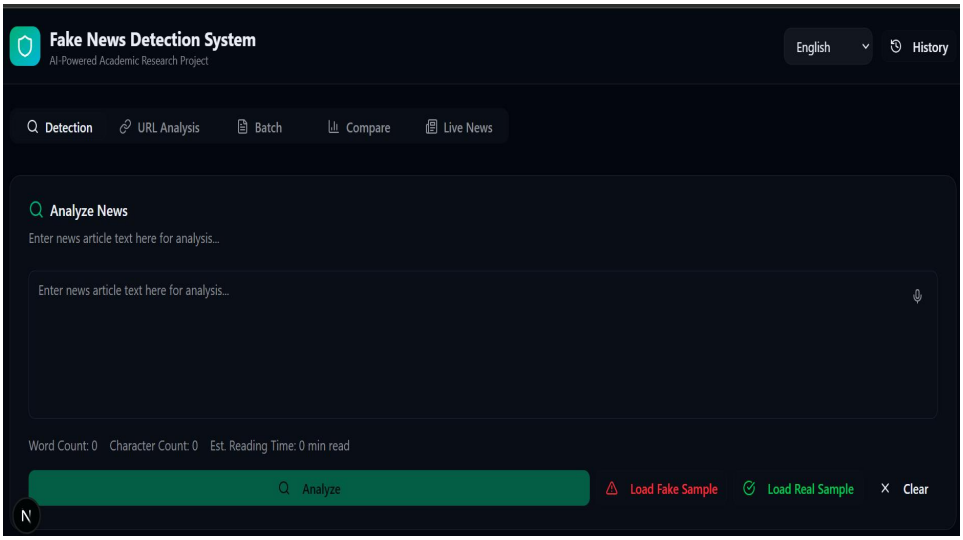
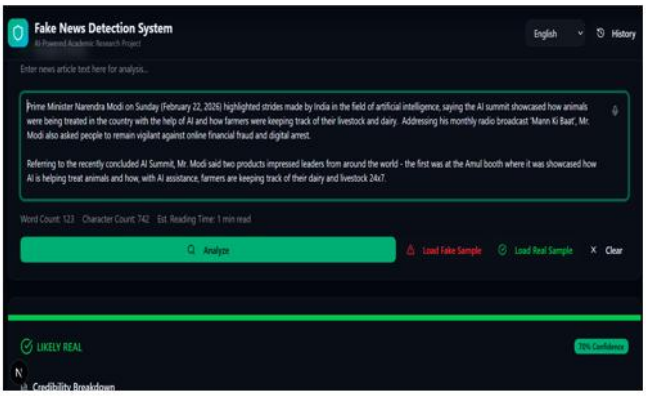
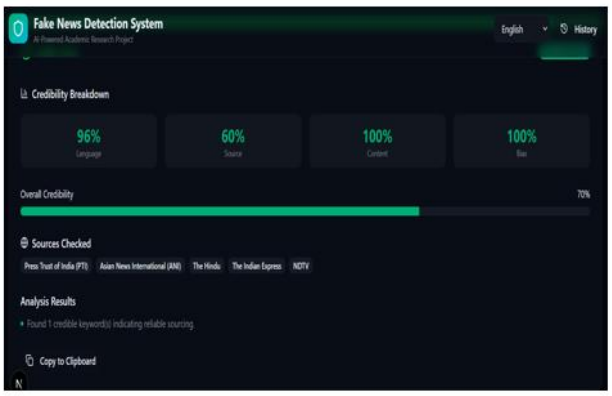


Fig. *Prototype Design*

Figure above shows the design of the prototype, which combines the Streamlit frontend interface with the preprocessing module, TF-IDF vectorizer, trained classification model, and fact-check API in a single backend processing system.





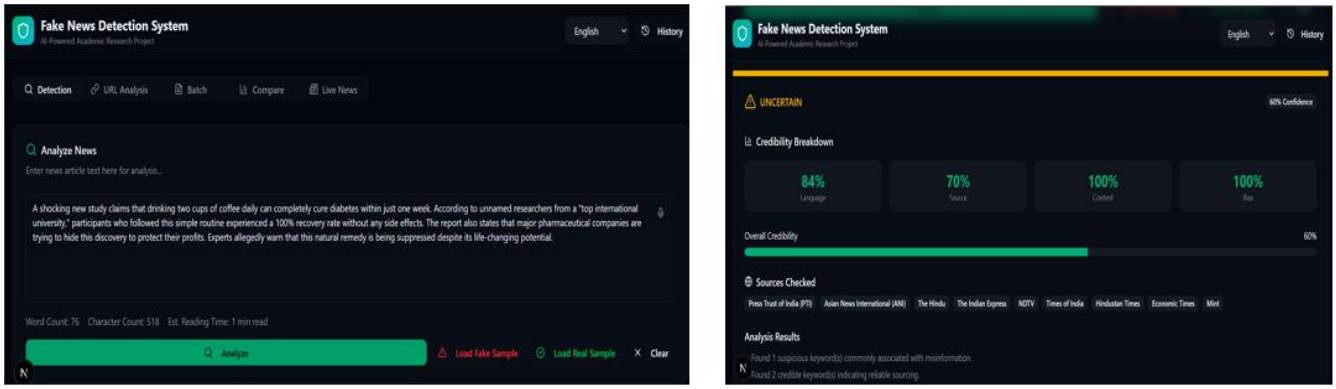


Fig. 16 *Working*

On the other hand, Figure 16 below shows how the prototype works through the use of real-time output screens. For example, when the authentic news content is fed into the system, it displays "LIKELY REAL" with a high confidence level and credibility percentage metrics such as language quality, source credibility, content consistency, and bias analysis.

The comparative analysis of various methods of news verification is shown in Table 4, which includes the Proposed Fake News Detection System, Manual Fact-Checking Platforms, Basic Keyword-Based Tools, and AI-Based Detection Systems. The various methods of news verification use different approaches to detect fake news. The manual platforms use human verification, which may result in a delay in the response. The keyword-based tools are rule-based and lack any understanding of the context. The AI-based systems use machine learning or deep learning models for automatic classification. The proposed system is novel in the sense that it uses Natural Language Processing, TF-IDF feature extraction, supervised machine learning models, and real-time API verification.

The proposed system is unique in that it combines Natural Language Processing, TF-IDF feature extraction, supervised machine learning models, and real-time API verification. The proposed system offers real-time predictions with confidence scores and credibility scores. In comparison to other systems, the proposed system is fully automated, scalable, and cost-effective for efficient fake news detection.

Features	Proposed System	Manual Fact-Checking	Basic Keyword Tools	AI-Based Tools
Detection Method	Ml+TF-IDF	Human Verification	Limited	ML Based
Real-Time Output	Yes	No	No	Yes
Confidence Score	Yes	No	No	Limited
Automation Level	Fully Automated	Semi-Automated	Automated	Automated
Technology Used	NLP,TF-IDF,ML Models	Human Review	Rule-Based	Deep Learning
Cost	Low	High	Low	Medium

Table . *Comparison of Fake News Detection System with Available Approaches.*

Note: Our understanding of evaluation methodologies and analytics techniques was cultivated through the Natutal Language Processing (CSE18R257). Leveraging this knowledge, we can systematically assess to ensure that our prototype algorithm meets the desired objectives and performs optimally in real-world scenarios.

CHAPTER 8

CONCLUSION & FUTURE SCOPE

This project entailed the design and implementation of a machine learning system for the detection of fake news using Natural Language Processing. The incorporation of different preprocessing methods and the application of TF-IDF to extract features from the text helped the system to effectively represent the unstructured news in a significant numerical manner. Different supervised learning models were trained and tested for the task of classifying real and fake news. The experiment results based on accuracy, precision, recall, and F1-score indicated that the proposed system is capable of offering accurate results. The inclusion of the trained model in the user-friendly interface makes the system even more helpful.

However, since the model has already achieved excellent results, there is still scope for improvement. Future studies may include the application of transformer models such as BERT to understand the context better and improve the detection of subtle misinformation. The system can be extended to process multilingual information, which will be more helpful for various regions. The system can also be extended to perform multimodal analysis by incorporating text, image, and metadata information. The system can also be developed to act as a browser extension or a social media monitoring system to make it more accessible. The system can also be updated periodically to keep up with the latest strategies of misinformation.

REFERENCES

1. F. Olan, U. Jayawickrama, E. O. Arakpogun, et al., “Fake news on social media: The impact on society,” Information Systems Frontiers, 2024.
2. E. Broda and J. Strömbäck, “Misinformation, disinformation, and fake news: Lessons from an interdisciplinary, systematic literature review,” Annals of the International Communication Association, 2024.
3. J. Alghamdi, S. Luo, and Y. Lin, “A comprehensive survey on machine learning approaches for fake news detection,” Multimedia Tools and Applications, 2024.
4. E. Papageorgiou, C. Chronis, I. Varlamis, and Y. Himeur, “A survey on the use of large language models (LLMs) in fake news,” Future Internet, 2024.
5. S. Baribi-Bartov, B. Swire-Thompson, and N. Grinberg, “Supersharers of fake news on Twitter,” Science, 2024.
6. F. Farhangian, R. M. O. Cruz, and G. D. C. Cavalcanti, “Fake news detection: Taxonomy and comparative study,” Information Fusion, 2024.
7. B. Hu, Q. Sheng, J. Cao, Y. Shi, Y. Li, D. Wang, et al., “Bad actor, good advisor: Exploring the role of large language models in fake news detection,” in Proc. AAAI Conf. Artificial Intelligence, 2024.
8. L. Peng, S. Jian, Z. Kan, L. Qiao, and D. Li, “Not all fake news is semantically similar: Contextual semantic representation learning for multimodal fake news detection,” Information Processing & Management, 2024.
9. J. O. Weatherall and C. O'Connor, “Fake News!,” Philosophy Compass, 2024.
10. J. Wu, J. Guo, and B. Hooi, “Fake news in sheep's clothing: Robust fake news detection against LLM-empowered style attacks,” in Proc. 30th ACM SIGKDD Conf., 2024.
11. R. Agrawal and P. Tiwari, “Hybrid ensemble models for fake news detection using NLP and deep learning,” International Journal of Computer Applications, vol. 178, no. 44, pp. 25–31, 2025.
12. S. P. Kumar, A. Sharma, and V. Ramesh, “Fake news classification using BERT and BiLSTM,” in Proc. 12th Int. Conf. on Machine Intelligence and Signal Processing, pp. 45–50, 2025.
13. N. Kale and M. K. Barbhuiya, “Real-time detection of fake news using transformers and fact-checking APIs,” Journal of Information Security Research, vol. 14, no. 1, pp. 10–18, 2025.
14. A. Rahman, T. Zaman, and H. Bhuiyan, “Evaluating multimodal fake news detection using deep fusion models,” IEEE Access, vol. 13, pp. 22250–22260, 2025.

15. D. Kapoor and S. Garg, “Low-resource fake news detection using few-shot learning,” in Proc. Int. Conf. on AI and Knowledge Engineering, pp. 98–104, 2025.

16. M. Srinivas and L. J. Reddy, “Fake news detection using CNN-LSTM hybrid model,” International Journal of Advanced Computer Science and Applications, vol. 16, no. 3, pp. 145–152, 2025.
17. Y. Chen, Q. Lu, and K. Zhang, “An explainable fake news detection approach based on attention mechanisms,” Expert Systems with Applications, vol. 237, pp. 120043, 2025.
18. K. A. Khan and R. S. Rao, “Integrating sentiment analysis with fake news detection for improved accuracy,” International Journal of Data Science and Analytics, vol. 9, no. 2, pp. 101–112, 2025.
19. A. B. Rathi, N. Gupta, and P. Verma, “Lightweight transformer models for fake news classification on mobile devices,” IEEE Transactions on Mobile Computing, vol. 24, no. 5, pp. 1901–1913, 2025.
20. P. Sundaram and S. T. Thomas, “A real-time AI-powered fake news monitoring system using hybrid architecture,” in Proc. IEEE Int. Conf. on Data Science and Artificial Intelligence, pp. 221–227, 2025.

KALASALINGAM
ACADEMY OF RESEARCH AND EDUCATION
(DEEMED TO BE UNIVERSITY)
Under sec. 3 of UGC Act 1956. Accredited by NAAC with "A++" Grade

INTERNAL QUALITY ASSURANCE CELL

PROJECT AUDIT REPORT

This is to certify that the project work entitled "Text-Based Fake News Detection Using Natural Language Processing and Machine Learning Approaches" categorized as an internal project done by L Jayadev Karthik, B Navyasree, P Bindu Pradhyumna and K Saketh Sreeram of the Department of Computer Science and Engineering, under the guidance of **M.K.Nagarajan** during the Even semester of the academic year 2025- 2026 are as per the quality guidelines specified by IQAC.

Quality Grade

Deputy Director (IQAC)

Administrative Quality Assurance **Director (IQAC)**